HDOC DAY-75

ENUMERATIONS IN C

Programming in C - Activity Submission

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First Semester

Question 1

Problem Statement: Assign explicit values starting from 10 in an enumeration and print them. Input: Nil. Expected Output: A=10, B=11, C=12.

Aim

To understand enumeration in C by assigning an explicit starting value and observing the automatically generated consecutive values.

Concept / Understanding

An enumeration (enum) is a user-defined data type in C that gives meaningful names to integer constants. When the first member is assigned 10, the next unassigned members automatically receive 11, 12, and so on.

Algorithm

1. Start the program.
2. Declare an enumeration containing A, B and C, and assign A = 10.
3. Print the integer values of A, B and C using printf().
4. End the program.

C Program

```
#include <stdio.h>

enum Values
{
    A = 10,
    B,
    C
};

int main()
{
    printf("A = %d\n", A);
    printf("B = %d\n", B);
    printf("C = %d\n", C);

    return 0;
}
```

Output

```
A = 10
B = 11
C = 12
```

Result

The program was executed successfully. A was assigned 10 explicitly, while B and C automatically received the consecutive values 11 and 12.

Question 2

Problem Statement: Use enumeration to represent menu choices (ADD, SUBTRACT, MULTIPLY) and perform the selected arithmetic operation using switch. Example: ADD 10 20 gives 30.

Aim

To use enumeration constants as menu choices and perform arithmetic operations using a switch statement.

Concept / Understanding

The enum constants ADD, SUBTRACT and MULTIPLY are assigned the values 1, 2 and 3. The user selects an option and enters two numbers. A switch statement then performs the corresponding operation.

Algorithm

1. Start the program.
2. Declare an enumeration Operation with ADD = 1, SUBTRACT and MULTIPLY.
3. Display the three menu choices.
4. Read the user choice and two integer values.
5. Use switch to perform addition, subtraction or multiplication.
6. Display the result; show an error message for an invalid choice.
7. End the program.

C Program

```
#include <stdio.h>

enum Operation
{
    ADD = 1,
    SUBTRACT,
    MULTIPLY
};

int main()
{
    int choice, num1, num2;

    printf("1. ADD\n");
    printf("2. SUBTRACT\n");
    printf("3. MULTIPLY\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter two numbers: ");
    scanf("%d %d", &num1, &num2);

    switch (choice)
    {
        case ADD:
            printf("Result = %d\n", num1 + num2);
            break;
        case SUBTRACT:
            printf("Result = %d\n", num1 - num2);
            break;
        case MULTIPLY:
            printf("Result = %d\n", num1 * num2);
            break;
```

```
        default:
            printf("Invalid choice\n");
    }

    return 0;
}
```

Sample Input

```
Enter your choice: 1
Enter two numbers: 10 20
```

Sample Output

```
Result = 30
```

Result

The program successfully uses enumeration constants with a switch statement to perform the selected arithmetic operation. For ADD with inputs 10 and 20, the output is 30.